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Earth Embeddings as Products

Taxonomy, Ecosystem, and Standardized Access

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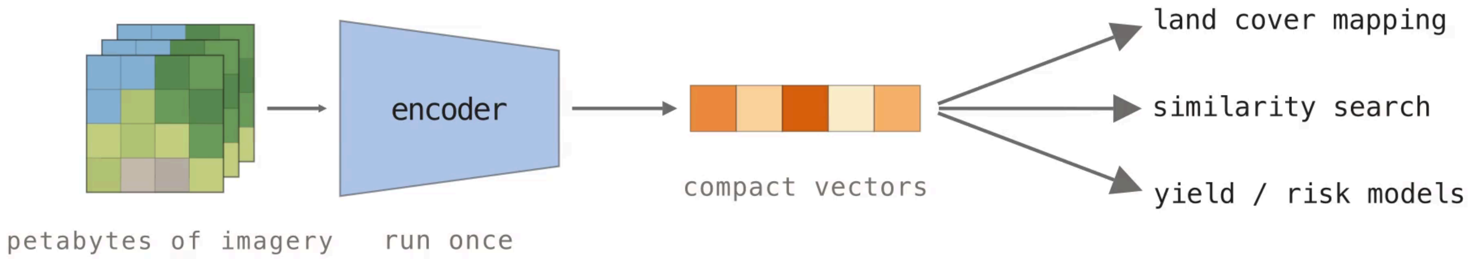
Embedding products move model inference out of the user’s workflow

Public EO archives are already petabyte-scale — NASA’s EOSDIS holds **178.7 PB** and grows by 160 TB per day. Foundation models can summarize this data, but every user re-pays the preprocessing and GPU bill. Pre-computed embedding products run the model once and ship the vectors as reusable data.



NASA ESDS annual metrics, FY2025 · Bommasani et al., 2021 — foundation model framing.

A foundation model is not an embedding product

Foundation models — dynamic inference

Distributed as public weights (DOFA, OlmoEarth). Users manage preprocessing and GPU inference themselves: flexible, but a hardware and engineering barrier.

Embedding products — static data

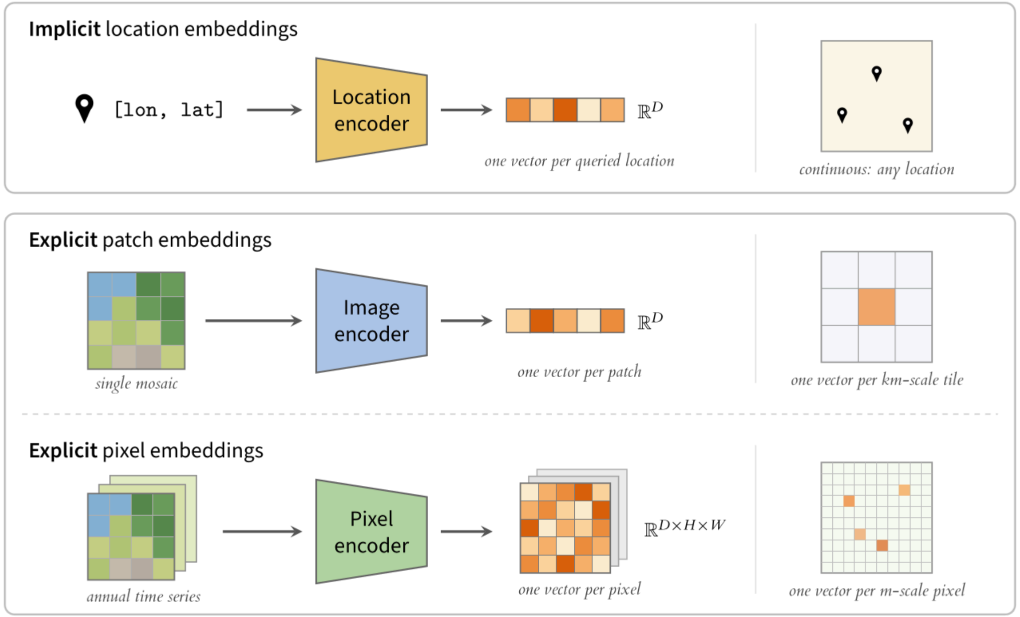
Distributed as pre-computed vector archives (Google Satellite Embedding). Frozen, versioned assets: analysis proceeds without the model or its compute.

The distinction is practical, not academic: a product is pinned to the data snapshot it was computed on, so **treating a product as a proxy for its model invites invalid generalization claims**.

Fang, Stewart, Corley, Zhu, Azizpour — Earth Embeddings as Products, IGARSS 2026, §II.

TAXONOMY

Three families: location, patch, and pixel embeddings



Stewart, Fang, Corley, Zhu — Earth Embeddings (book chapter, 2026), Fig. 1; figure design adapted from Klemmer et al., 2025.

Patch products: km-scale vectors built for search

All known patch embedding products as of July 2026. *sparse spatial or temporal coverage.

Product	Extent	Resolution	Years	Dim.	Dtype
MOSAIKS (2021–25)	USA / Global	1 km / 0.01°	2018 / 2019	8192 / 4000	float32
Clay v0 / v1.5 (2024–26)	USA / Global	154 m – 5.12 km	2010–2025*	768–1024	float32
Major TOM (2024–26)	Global*	120 m – 3.84 km	2016–2024*	256–2048	float32
Earth Index (2025)	Global	320 m	2024	384	float32
Copernicus-Embed (2025)	Global	0.25°	2021	768	float32

Major TOM is the largest family — 14+ model variants over one shared grid. Every product stores float32; one vector summarizes an entire mosaic tile, which suits retrieval more than dense mapping.

Earth Embeddings (book chapter, 2026), Table 2 — abridged to family level.

Pixel products: dense annual vector maps

All known pixel embedding products as of July 2026 — every one is built from annual time series. *sparse coverage.

Product	Extent	Resolution	Years	Dim.	Dtype
Presto Embeddings (2025)	Togo	10 m	2019–2020	128	uint16
Tessera Embeddings (2025)	Global*	10 m	2017–2025*	128	int8 → float32
Google Satellite Embedding (2025)	Global	10 m	2017–2025	64	int8 → float64
Embedded Seamless Data (2026)	Global	30 m	2000–2024	12	uint16 → float32

Google Satellite Embedding covers the full Sentinel era at just 64 dimensions; **Embedded Seamless Data** trades resolution for 25 years of Landsat history. Storage pressure drives the int8/uint16 dtypes.

Earth Embeddings (book chapter, 2026), Table 3.

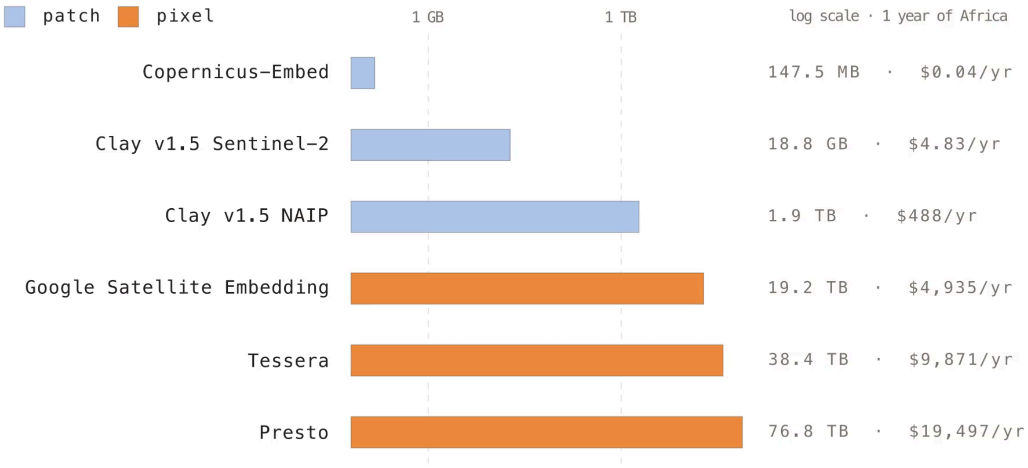
Every product ships differently, and users pay for it

- **Distribution is scattered** — Source Cooperative (Clay, Earth Index), Hugging Face (Major TOM, Copernicus-Embed), Google Earth Engine (Google, Presto), and private university servers (Tessera).
- **Formats disagree** — GeoParquet, GeoTIFF with implicit CRS assumptions, and raw NumPy arrays with no CRS or bounds: "numbers and a prayer."
- **Grids are proprietary** — Major TOM’s grid vs. MGRS vs. per-product tiling, so cross-product comparison starts with reprojection.
- **One flipped coordinate** in the Google Satellite Embedding rasters required patches to **GDAL, rasterio, and TorchGeo** before standard tools could read them.

Every team solves distribution independently — the integration tax is paid once per product, per user.

Corley — The Technical Debt of Earth Embedding Products, cloudnativegeo.org, Feb 2026.

Storage for one continent-year spans five orders of magnitude



Patch products stay in the MB–GB range; dense 10 m pixel products reach tens of TB. A single full download of Presto's Africa coverage adds ~**$6.9k in egress** on top of storage.

Earth Embeddings (book chapter, 2026), Table 8 — AWS S3 Standard, us-west-2 · egress from Corley, 2026.

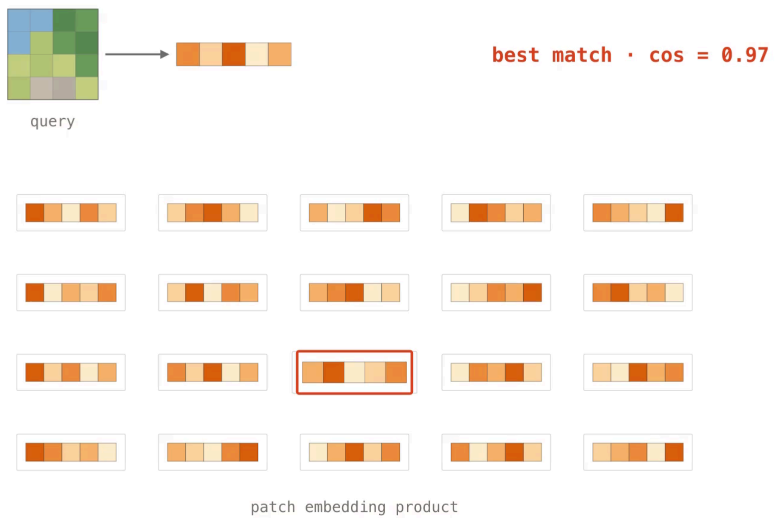
Patch products are reproducible; pixel models often are not

- **Clay, Earth Index, Copernicus-Embed** — the full pipeline (code, weights, data, embeddings) is released under permissive licenses.
- **Major TOM** — CC-BY-SA pretraining data makes every derived embedding copyleft, which keeps many commercial users away.
- **AlphaEarth Foundations and ESDNet** — proprietary code and weights. The embeddings are CC-BY, but no one outside can regenerate, audit, or improve them.
- **No product ships checksummed data** — upstream archives keep reprocessing, so bit-for-bit reproduction of pretraining or inference inputs is already impossible.

Earth Embeddings (book chapter, 2026), Tables 4–6 — license provenance from data to weights to embeddings.

TorchGeo treats embeddings as first-class geospatial datasets

- Loaders for **every known embedding product:** Clay, Major TOM, Earth Index, Copernicus-Embed, Presto, Tessera, Google.
- The generating **models and weights** too — Presto and Tessera added upstream.
- Reprojection, rasterization, intersection, and samplers come for free.
- **4+ repositories** of custom glue become ~20 lines against one API.



Stewart et al., 2022 — TorchGeo · Fang et al., IGARSS 2026, §IV.

Two workflows, one API: retrieval and mapping

Search & retrieval — query by example

```
# torchgeo.datasets / torchgeo.models
earthindex = EarthIndexEmbeddings('data/ei')
s2 = Sentinel2('data/s2')
image = s2[xmin:xmax, ymin:ymax]['image'] / 10_000

model = vit_small_patch14_dinov2(
    ViTSmall14_DINOv2_Weights.SENTINEL2_ALL_SOFTCON)
embed = model(image)

cos = CosineSimilarity(dim=0)
for sample in iter(earthindex):
    sim = cos(embed, sample['embedding'])
    ...  # keep the argmax, plot the match
```

Land cover mapping — crop types across Europe

```
# torchgeo.datasets / torchgeo.samplers
tessera = TesseraEmbeddings('data/tessera')
eurocrops = EuroCrops('data/ec', download=True)
dataset = tessera & eurocrops  # intersection

train_roi = box(-10, 35, 10, 60)  # West EU
test_roi  = box( 10, 35, 30, 60)  # East EU
train_ds, test_ds = roi_split(
    dataset, [train_roi, test_roi])

# random patches → fit a k-NN / linear probe
# gridded patches → stitch a map of Europe
```

Earth Embeddings (book chapter, 2026), §4 Listings 1–2 — abridged; geographic train/test splits are built in.

Evidence so far: strong for mapping, weaker under transfer

Task	Embeddings	Finding
Cropland mapping, Togo	Presto, Google	Presto + Random Forest gives the best F1
Tree species, Dutch forest inventory	Presto, Google, Tessera	+2–9 points over hand-designed time-series features
Landslide susceptibility, TW·HK·IT	Google	64-d embeddings beat conventional conditioning factors
Poverty mapping, Sub-Saharan Africa	Google + GNN	large storage/preprocessing savings vs. raw Sentinel-2
Scene classification, EuroSAT-Embed	Google, Tessera, OlmoEarth	pooling choice cuts the geographic gap by >50%
Fusion across six tasks	Google, Tessera, GeoCLIP, SatCLIP	fused embeddings beat the best single model in 4 of 6

Under **spatial transfer** performance degrades, and annual composites wash out sub-annual dynamics — validate with geographic splits, against simple baselines, before trusting any product.

Zvonkov et al. · Ishikawa et al., 2025 · Cheng et al., 2026 · Pettersson & Daoud, 2025 · Corley et al., 2026 · van der Plas et al., 2026 · Ma et al., 2026.

Publish embeddings like data products, not model dumps

Embedding type	Format
Location / implicit	ONNX or PyTorch
Patch / region	GeoParquet
Pixel, snapshot	GeoTIFF or GeoZarr
Pixel, time series	GeoZarr

Ship a model card: sensors, temporal window, CRS and grid, dtype, quantization transform, license — in the files, not the docs.

- **int8 quantization is near-free** — negligible accuracy loss; Google and Tessera already do it.
- **PCA to 64 dims + int8** gives 64× compression with <2% accuracy loss.
- **Binary quantization** adds another 32× and still recovers ~65% of float32 nearest neighbors — enough for candidate retrieval.
- Ship a **runnable benchmark**, not a private leaderboard.

Earth Embeddings (book chapter, 2026), §7 · Corley & Robinson, 2026 — TerraBit compression study.

TAKEAWAYS

Embeddings are usable today; standards are the remaining work

Pick the family by task — implicit for location context, patch for retrieval, pixel for dense mapping. Validate under geographic splits against simple baselines. Open problems: ocean and atmosphere coverage, uncertainty layers, and shared benchmarks — identical models differ by 10+ points across papers.

Chapter	Earth Embeddings — arXiv, August 2026
Survey	github.com/hfangcat/Awesome-Geospatial-Embeddings
Code	github.com/torchgeo/torchgeo
Contact	isaac.corley@taylorgeospatial.org